Tutorial 12

MAT3007 Optimization

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LP relaxation

- In general, IP is very hard to solve. We can relax all integral or binary constraints and solve it as a linear program.
- There are some cases in which solving the LP relaxation can give one very good solution.
- If the optimal solution to the LP relaxation is integral, then the solution must be optimal to the IP problem (Theorem).

Special LP relaxation: Totally Unimodularity

- Totally Unimodularity: A matrix A is said to be totally unimodular if the determinant of each submatrix of A is either 0, 1, or 1.
- If the constraint matrix A is totally unimodular and b is integral, then all the BFS are integers and the LP must have an optimal solution that is an integral solution.

Totally Unimodularity Example1

Recall the matching problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n} \sum_{j=1}^{n} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_{ij} = 1 \qquad \forall j \\ & \sum_{j=1}^{n} x_{ij} = 1 \qquad \forall i \\ & x_{ij} \in \{0, 1\} \end{aligned}$$

The constraint matrix is TU.
An example when $n = 2$ (in the order of $x_{11}, x_{12}, x_{21}, x_{22}$):

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

One can choose $B = \{1, 2\}$, $C = \{3, 4\}$. The condition holds and A is TU.

Totally Unimodularity Example2

Matrices that are not TUM:

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Matrices that are TUM:

$$\begin{pmatrix} 1 & -1 & 0 & 0 & -1 \\ -1 & 1 & -1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & -1 \\ -1 & 0 & 0 & -1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & -1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Totally Unimodularity Example3

Consider the following matrix A:

$$A = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{bmatrix}$$

Then we can make the partition to be $B = \{1, 2, 3, 4\}$, $C = \phi$. The condition holds and A is TU

- If the constraint matrix A is totally unimodular and b is integral, then all the BFS are integers and the LP must have an optimal solution that is an integral solution.
- However, TU property is uncommon in practice - if one solves the LP relaxation and find the optimal BFS is not integral, it means that TU doesn't hold

General Idea of Branch-and-Bound

- Divide the feasible region into several non-overlapping parts, solve the optimization problem on each part, compare the solution in each part, and find the optimal solution to the original problem.

General Idea of Branch-and-Bound

First, solve the LP relaxation of the IP:

- If the solution is already integral, then it is optimal to the IP

If the optimal solution to the LP relaxation is $\boldsymbol{x}^*$ and x_i^* is fractional, then branch the problem into the following two:

1. One with an added constraint that $x_i \leq \lfloor x_i^* \rfloor$, call this (S1)
2. One with an added constraint that $x_i \geq \lceil x_i^* \rceil$, call this (S2)

Get two IP's after branching.

- Then solve (S1) and (S2) and assume we can get the optimal solutions $\boldsymbol{x}_1^*$ and $\boldsymbol{x}_2^*$ with optimal values v_1^* and v_2^* (both (S1) and (S2) are still integer programs).

Claim

- *If $v_2^* \leq v_1^*$, then $\boldsymbol{x}_1^*$ is the optimal solution to the original IP*
- *If $v_1^* \leq v_2^*$, then $\boldsymbol{x}_2^*$ is the optimal solution to the original IP*

The claim is true because the union of the feasible regions in each branch equals the feasible region of the original problem

Complexity of Branch-and-Bound

- In the worst case, branch-and-bound may need to go through each feasible integer solutions in the region, which is exponential in the problem size.
- However, branch-and-bound does enumeration in a smart way and typically it only needs to visit a tiny fraction of all solutions

Exercise 1: Travelling salesman problem(TSP)

- TSP is an NP-hard problem in combinatorial optimization, important in theoretical computer science and operations research.
- Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?
- Decision variable: $X_{ij} = 1$ if the path goes from city i to city j, $X_{ij} = 0$ otherwise.
- Take $c_{ij} > 0$ to be the distance from city i to city j. Please write the integer linear programming for TSP.

Exercise 1 solution

TSP can be written as the following integer linear programming problem:

$$\min \sum_{i=1}^{n} \sum_{j \neq i, j=1}^{n} c_{ij} x_{ij}:$$

$$\sum_{i=1, i \neq j}^{n} x_{ij} = 1 \qquad j = 1, \ldots, n;$$

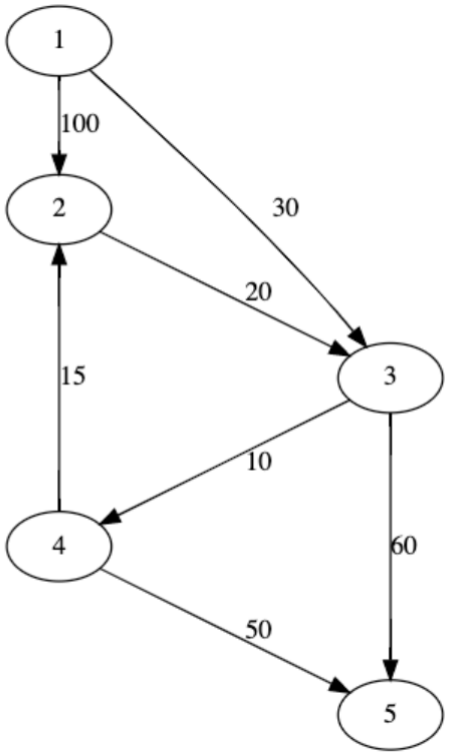
$$\sum_{j=1, j \neq i}^{n} x_{ij} = 1 \qquad i = 1, \ldots, n;$$

$$\sum_{i \in Q} \sum_{j \neq i, j \in Q} x_{ij} \leq |Q| - 1 \qquad \forall Q \subsetneq \{1, \ldots, n\}, |Q| \geq 2$$

The last constraint of the DFJ formulation—called a subtour elimination constraint—ensures no proper subset Q can form a sub-tour, so the solution returned is a single tour and not the union of smaller tours. Because this leads to an exponential number of possible constraints, in practice it is solved with row generation.

Exercise 2: Shortest path problem(SP)

- In graph theory, the shortest path problem is the problem of finding a path between two vertices in a graph such that the sum of the weights of its constituent edges is minimized.
- Decision variable: x_{ij} is an indicator variable for whether edge (i, j) is part of the shortest path: 1 when it is, and 0 if it is not. Take $a_{ij} > 0$ to be the cost from node i to node j.



Exercise 2 solution

SP can be written as the following integer linear programming problem:

$$\min \quad \sum_{\forall e(i,j) \in E} a_{i,j} \times x_{i,j}$$

$$s.t. \quad b(i) = \sum_j x_{ij} - \sum_k x_{ki} = \begin{cases} 1 & \text{if } i \text{ is the starting node,} \\ -1 & \text{if } i \text{ is the ending node,} \\ 0 & \text{otherwise.} \end{cases}$$

$$x_e \in \{0, 1\} \qquad \forall e \in E$$

We wish to select the set of edges with minimal weight, subject to the constraint that this set forms a path from the starting node s to the ending node t, represented by the equality constraint: for all vertices except s and t the number of incoming and outcoming edges that are part of the path must be the same.

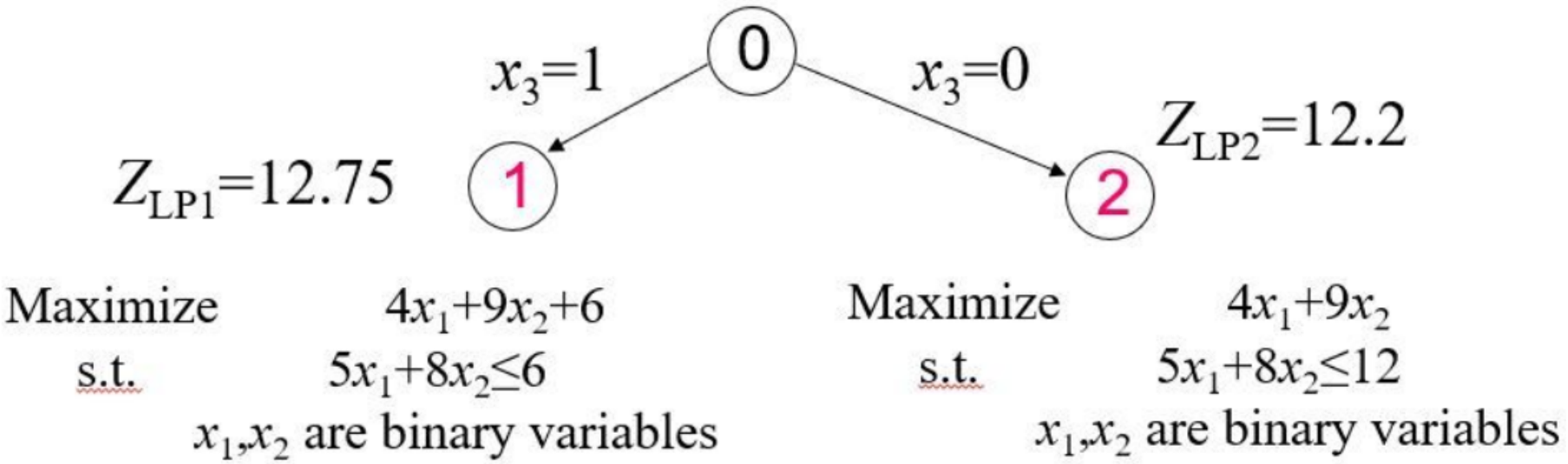
Exercise 3: Branch-and-Bound

Solving the following problem by Branch-and-Bound algorithm.

$$\begin{aligned} &\text{Max}_x \ 4x_1 + 9x_2 + 6x_3 \\ &\text{subject to } 5x_1 + 8x_2 + 6x_3 \leq 12 \\ &x_1, x_2, x_3 \in \{0, 1\} \end{aligned}$$

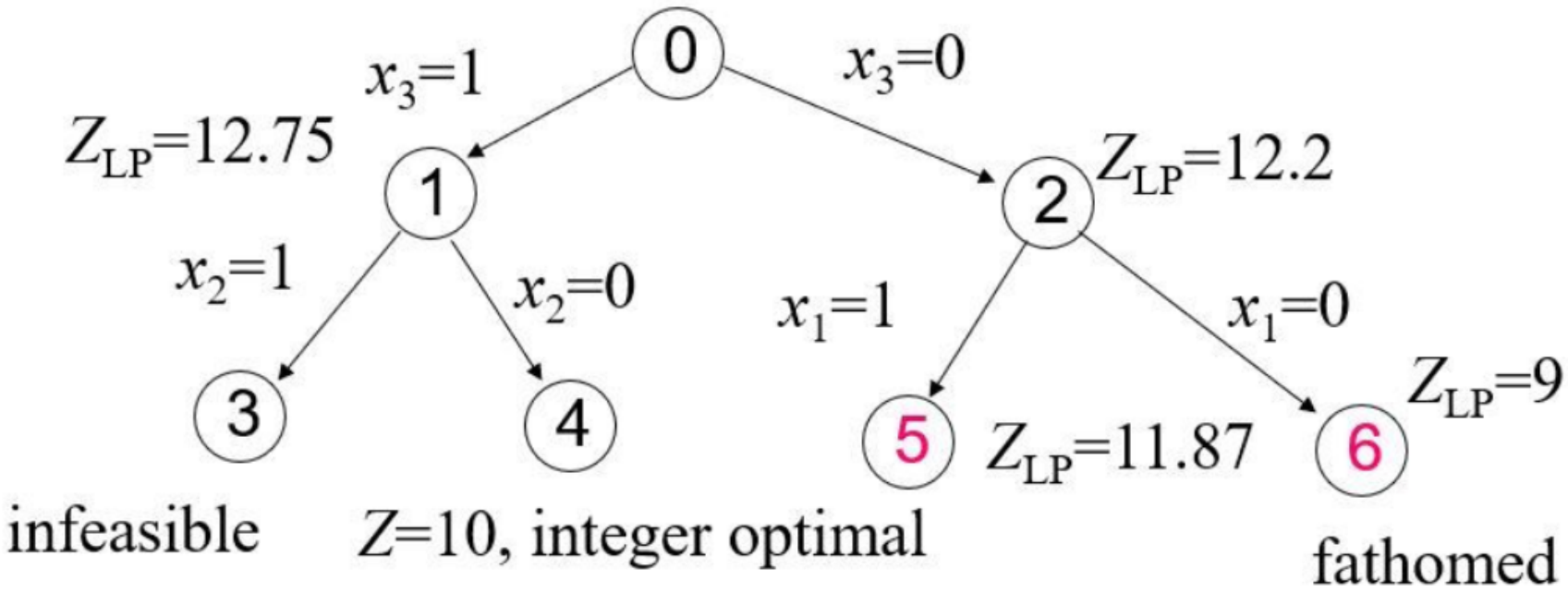
Exercise 3 solution

Step 1: Solving the LP relaxation of original problem and get the optimal value $x_1 = 0$, $x_2 = 1$, $x_3 = 0.667$. Then we need to consider two subproblems, one with $x_3 = 1$, the other with $x_3 = 0$.



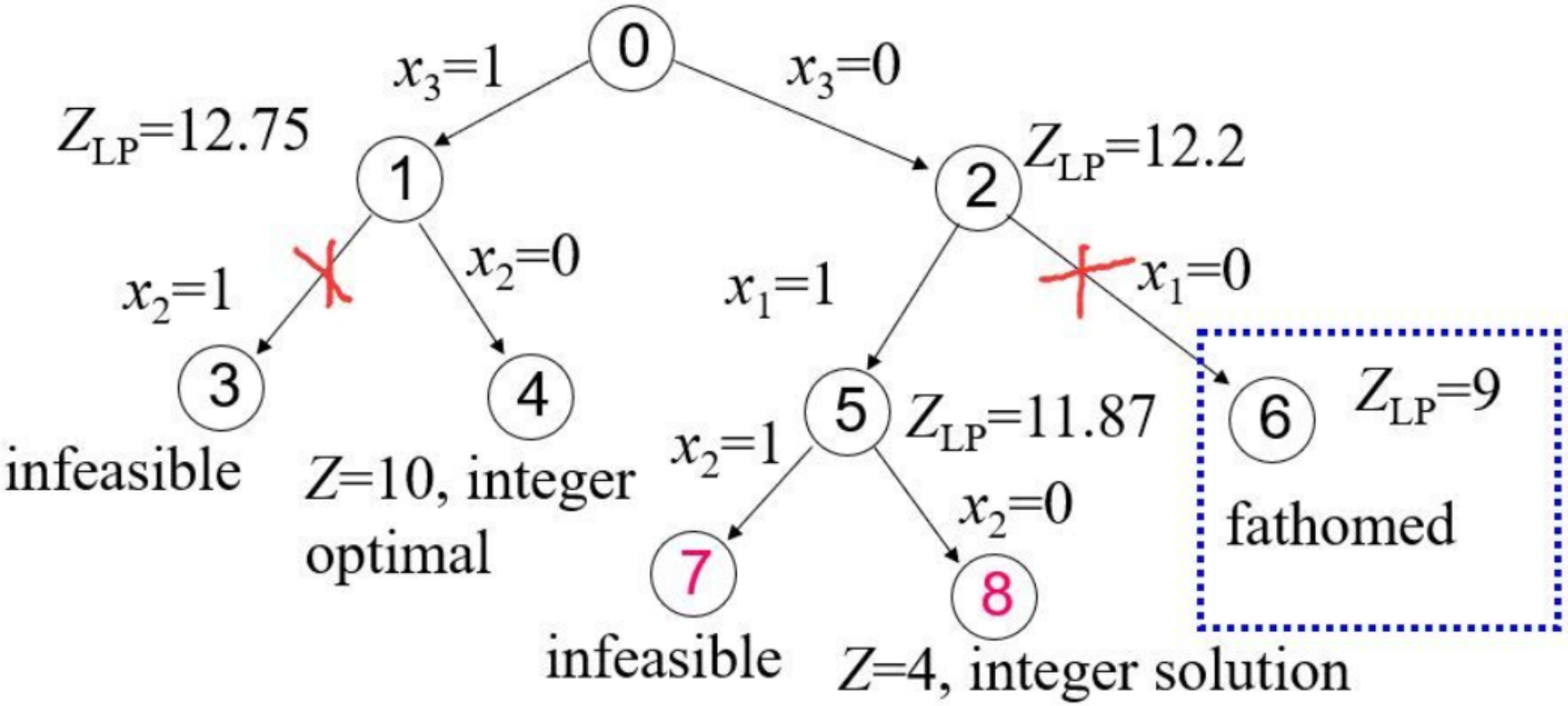
Exercise 3 solution

Step 2: Solving the subproblems(S1) and (S2). Both of their results are feasible. So we continue branching both subproblems. Subproblem(S3) is infeasible. The optimal value of (S4) is 10, we do not need to search for the subproblems of (S4). The optimal value of (S5) is 11.87. The optimal value of (S6) is 9, it is smaller than optimal value of (S4), so we abandon (S6).



Exercise 3 solution

Step 3: Solving the subproblems (S7) and (S8), we do not find better solution. So the optimal solution is 10.



Acknowledgement

Thank you for your time!